

Session 2 · Deck 2.A1

Bewley & Huggett: Why Can't Everyone Insure?

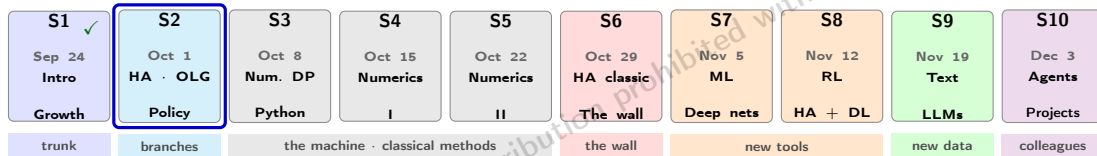
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Thursday, October 1, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.A1

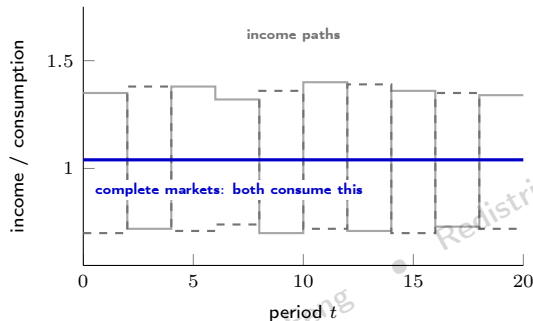


Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 2.A1 why insurance fails and what replaces it · **2.A2** Aiyagari: production and the capital market · **2.A3** the stationary distribution, and where the model is taken next.

A Prediction You Can Test

Two households, same lifetime income, different luck



Thin lines: two realizations of the same i.i.d. endowment process. Thick line: the full-insurance allocation.

The complete-markets prediction

With a full set of state-contingent claims, marginal utilities move *together*:

$$\frac{u'(c_{t+1}^i)}{u'(c_t^i)} = \frac{u'(c_{t+1}^j)}{u'(c_t^j)} \quad \text{for all } i, j.$$

Individual income is **irrelevant** for consumption growth; only aggregate risk is priced.

It fails in the data. Consumption responds to idiosyncratic income — more so for the young and the poor.

Cochrane (1991); Attanasio & Davis (1996); Blundell, Pistaferri & Preston (2008).

The Arrow–Debreu Benchmark: Model and First-Order Condition

Household i has endowment $y^i(s^t)$ at history $s^t = (s_0, \dots, s_t)$, of probability $\pi(s^t)$. At date 0 it buys claims to every history, at prices $p(s^t)$, under **one** budget constraint:

$$\max_{\{c^i(s^t)\}} \sum_t \sum_{s^t} \beta^t \pi(s^t) u(c^i(s^t)) \quad \text{s.t.}$$

$$\sum_t \sum_{s^t} p(s^t) c^i(s^t) \leq \sum_t \sum_{s^t} p(s^t) y^i(s^t);$$

One constraint, one multiplier μ^i ; the first-order condition:

$$\beta^t \pi(s^t) u'(c^i(s^t)) = \mu^i p(s^t).$$

Prices clear markets: $\sum_i c^i(s^t) = \sum_i y^i(s^t) \equiv Y(s^t)$ at every history, so with $p(s_0) = 1$, $p(s^t) = \beta^t \pi(s^t) u'(Y(s^t)) / u'(Y_0)$.

Full insurance in two lines

Divide i 's condition by j 's. The price and the probability cancel:

$$\frac{u'(c^i(s^t))}{u'(c^j(s^t))} = \frac{\mu^i}{\mu^j} \quad \text{for every } t, s^t.$$

With CRRA(γ) utility, $c^i(s^t) = \theta^i Y(s^t)$: a **fixed share** of the aggregate. Household i 's own endowment enters only through $\theta^i = \sum p y^i / \sum p Y$, its share of wealth at these prices.

Planner: weights $\omega^i \propto 1/\mu^i$ give the same allocation, with $p(s^t)$ as the multiplier on resources (Negishi). Shares are not weights: $\theta^i \propto (\omega^i)^{1/\gamma}$.

The Test, and Why It Fails

Take logs of $c_t^i = \theta^i C_t$ and difference:

$\Delta \log c_t^i = \Delta \log C_t$. That gives a regression anyone can run on panel data:

$$\Delta \log c_t^i = \alpha \Delta \log C_t + \zeta \Delta \log y_t^i + \varepsilon_t^i.$$

Complete markets predict $\alpha = 1$ and $\zeta = 0$: once aggregate consumption is controlled for, your own income should not matter.

- **Rejected.** Consumption falls after long illness and involuntary job loss (Cochrane 1991), and tracks relative wages across cohorts (Attanasio & Davis 1996). Villages come close but not all the way (Townsend 1994; Mace 1991).
- **Partial, not absent.** Transitory income shocks are largely smoothed; permanent ones largely pass through (Blundell, Pistaferri & Preston 2008).

Neither extreme

The data sit between **full insurance** ($\zeta = 0$) and **autarky** ($c = y$). A representative agent cannot produce that middle ground.

Why a heterogeneous-agent model

Give households *one* asset and let them self-insure. Savings absorb transitory shocks; a permanent shock cannot be saved away. That is the pattern in the data — and it reproduces much of it quantitatively. Kaplan & Violante (2010).

The price: the wealth distribution becomes part of the equilibrium.

Complete vs. Incomplete Markets: What Changes

	Complete markets	Incomplete markets (one bond)
Consumption	fixed share of C_t ; own income irrelevant	moves with own income and wealth
Wealth distribution	pinned by initial wealth; irrelevant for aggregates	non-degenerate, and a state variable
Interest rate	$r = \rho \equiv 1/\beta - 1$ in a steady state	$r^* < \rho$: precautionary demand for assets
MPC out of a transfer	small and equal for everyone	near 1 at the constraint, near 0 for the rich
Aggregation	a representative agent exists	the distribution matters for prices
Welfare theorems	hold: solve the planner instead	fail: solve the equilibrium directly
Transfers, public debt	neutral	real effects on saving, r and insurance
Risk aversion	prices <i>aggregate</i> risk only	drives saving, r^* , the wealth distribution and the welfare cost of risk

One missing set of contracts changes **every** row. The last row is why preferences matter more here: under complete markets idiosyncratic risk is traded away, so the curvature of u never meets it; under incomplete markets households bear it, and how much they *mind* it decides how much they save.

Where the Insurance Goes Missing

The Arrow–Debreu benchmark needs a claim to every history, and needs every claim honoured.

Take that away. Three frictions, any one of which suffices:

- **Missing markets** — only a risk-free claim trades.
- **Limited commitment** — contracts cannot be enforced.
- **Private information** — the shock is unobserved.

This course takes the first road

We *assume* the asset menu: one risk-free bond, plus a borrowing limit. This is the **Bewley–Huggett–Aiyagari** tradition.

It is assumed, not derived, because it is tractable, disciplined by data, and computable.

Research in Practice

The friction is a *modelling choice*, not a theorem. A welfare number from such a model inherits that choice — state it when you report one.

The Other Roads: Incompleteness Derived, Not Assumed

Limited commitment. Any household can walk away from an insurance contract and live on its own endowment. So every contract must be worth at least **autarky**, in *every* state.

- The lucky household is the one asked to pay. Ask too much and it defaults — so insurance is **partial**, and consumption rises when a participation constraint binds.
- Decentralized, this is a **state-contingent borrowing limit** “not too tight”: the largest debt a household still prefers to repay.

Kehoe & Levine (1993); Kocherlakota (1996); Alvarez & Jermann (2000).

Private information. If only the household sees its income, it will report bad luck. To keep reports truthful, consumption must **rise with reported income** — insurance again partial. Green (1987); Thomas & Worrall (1990); Atkeson & Lucas (1992).

The common intuition

Insurance transfers resources from the lucky to the unlucky. Both frictions put a limit on how much the lucky can be made to give up: one because they can leave, the other because they can hide.

So the *amount* of insurance is an equilibrium outcome, and it **responds** to policy and to the risk itself (Krueger & Perri 2006).

Research in Practice

The cost is computational: no planner's problem, and a recursive equilibrium on (a, e) may not exist — you need an **expanded state space** whose fixed point is the *set* of equilibria. Feng, Miao, Peralta-Alva & Santos (IER 2014) — 1.A2's warning made concrete; the device returns in 2.B3.

Huggett (1993): The Simplest Economy That Works

- A continuum of households, unit mass. **No production:** each receives an endowment $e_t \in \mathcal{E}$, a finite Markov chain with transition $\pi(e, e')$.
- One asset: a one-period discount bond, price q , paying one unit next period. Holdings a , with $a' \geq -\underline{a}$.
- The household solves

$$v(a, e) = \max_{c, a'} \left\{ u(c) + \beta \sum_{e'} v(a', e') \pi(e, e') \right\}$$

subject to $c + q a' = a + e$, $a' \geq -\underline{a}$, $c \geq 0$.

- **Zero net supply:** the bond is an IOU between households. In equilibrium

$$\int a'(a, e) d\lambda(a, e) = 0.$$

Why start here

No capital, no firms, no wage. The **only** price is q , and the only equilibrium condition that bites is that borrowing equals lending.

Everything hard about heterogeneous agents is already present: a cross-sectional distribution λ , a price that depends on it, and no planner to appeal to.

$\lambda(a, e)$ is a **probability measure** over (a, e) , not a list of names. Households are anonymous; only the distribution matters for prices.

The Borrowing Limit Is Not a Free Parameter

Natural limit. The largest debt a household can *always* repay, even drawing the worst endowment forever:

$$\underline{a}^{\text{nat}} = \frac{y_{\min}}{r},$$

which in the Aiyagari economy of 2.A2 is $w\epsilon_{\min}/r$.

Ad hoc limit. Any $\underline{a} \leq \underline{a}^{\text{nat}}$, imposed to match observed debt or to stand in for unmodelled default risk.

- At the natural limit it never binds — $c \rightarrow 0$ is avoided only because $u'(0) = \infty$.
- Tighter limits make it bind for a positive mass.

That mass is what policy questions care about.

Always $a' \geq -\underline{a}$ with $\underline{a} \geq 0$ here; the sister-course slides flip this sign.

A modelling decision with teeth

The constrained fraction drives the consumption response to a transfer and the optimal level of social insurance. In most papers the limit is *chosen*, not estimated.

In your own work: say whether $\underline{a}$ was imposed or estimated, and show how the results move when it changes.

Or derived, not assumed

On the limited-commitment road of two slides back, $\underline{a}$ is not a parameter but an **equilibrium object**: the largest debt a household still prefers to repay.

The Household's Optimality Conditions

Attach a multiplier η to the borrowing constraint. The **Kuhn–Tucker** conditions of the household's problem are the Euler equation

$$q u'(c) - \beta \mathbb{E}[u'(c') | e] = \eta$$

together with **complementary slackness**,

$$\eta \geq 0, \quad a' + \underline{a} \geq 0, \quad \eta (a' + \underline{a}) = 0.$$

Those three conditions split the problem in two:

- $a' > -\underline{a} \Rightarrow \eta = 0$: the Euler equation holds **with equality**.
- $a' = -\underline{a} \Rightarrow \eta > 0$: $q u'(c) > \beta \mathbb{E}[u'(c') | e]$ — the household *would* borrow more, and cannot.

u is concave on a convex budget set, so these are necessary **and** sufficient.

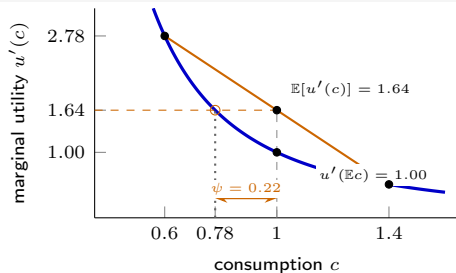
η is what the solver chases

The multiplier is not a bookkeeping device. It is the **shadow price of the borrowing limit** — how much a household would pay to be allowed one more unit of debt.

It is zero for most households and strictly positive for the ones every policy question is about.

And it is why the problem is awkward to compute: the system is a set of *inequalities*, and which branch applies is itself unknown.

Precautionary Saving I: Where the Extra Assets Come From



$u'(c) = c^{-2}$; $c \in \{0.6, 1.4\}$ equally likely, so $\mathbb{E}c = 1$. The chord lies above the curve: Jensen.

1. **No state-contingent claim.** The bond moves resources across *time*, but pays the same in every state. So c' still varies with next period's shock.
2. **Convex marginal utility.** A spread in c' raises the *average* of u' : $\mathbb{E}[u'(c')] = 1.64 > 1.00 = u'(\mathbb{E}c')$.
3. **The Euler equation** $q u'(c) = \beta \mathbb{E}[u'(c')] + \eta$ now demands a **higher** $u'(c)$ today: lower c , more saving, at *every* q .

Precautionary premium (Kimball 1990):

$u'(\mathbb{E}c' - \psi) = \mathbb{E}[u'(c')]$. Here c' counts as a sure $1.64^{-1/2} = 0.78$: $\psi = 0.22$ (arrow).

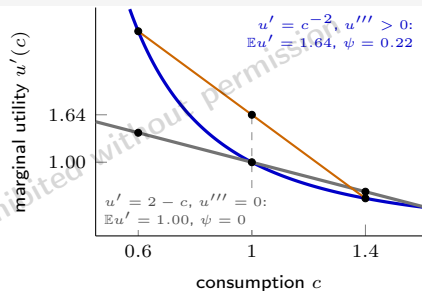
Under complete markets link 1 is cut: c' does not depend on one's own shock, so there is nothing for convexity to act on. **Incompleteness creates the risk; the shape of u' decides whether it raises saving** — next slide.

Precautionary Saving II: Prudence, Not Risk Aversion Alone

Two different curvatures, two different questions (Kimball 1990):

	Risk aversion	Prudence
Measure	$A = -u''/u'$	$P = -u'''/u''$
Premium	$\pi \approx \frac{1}{2}A\sigma^2$	$\psi \approx \frac{1}{2}P\sigma^2$
Answers	pay to <i>remove</i> risk?	save against risk?

- **Example** ($u' = c^{-2}$, $\sigma^2 = 0.16$): $A = 2$, $\pi \approx 0.16$; $P = 3$, $\psi \approx 0.24$ (exact 0.22).
- **Plausible case:** $A'(c) = A(A - P)$, so A falls with wealth iff $P > A$. CRRA: $\gamma + 1 > \gamma$. Quadratic: A rises.
- **Second motive:** a limit that may bind *later* makes even a $u''' = 0$ household save, with $\eta = 0$ today.
- **Risk aversion is necessary, not sufficient:** precautionary saving needs $u''' > 0$.



Same lottery $c' \in \{0.6, 1.4\}$; both utilities are risk averse and have $u'(1) = 1$. On the straight line the chord *is* the line: risk leaves saving unchanged (certainty equivalence), even with incomplete markets.

Leland (1968); Sandmo (1970); Kimball (1990); Carroll & Kimball, "Liquidity Constraints and Precautionary Saving," NBER WP 8496 (2001).

One Equation Instead of Three Conditions

Those three conditions are a **complementarity problem**:

$$0 \leq \eta \perp (a' + \underline{a}) \geq 0.$$

A root-finder wants *equations* and a gradient wants *smoothness*; inequalities give neither. The **Fischer–Burmeister** function turns all three into one:

$$\phi(x, y) = x + y - \sqrt{x^2 + y^2},$$

which vanishes *exactly* when $x \geq 0$, $y \geq 0$ and $xy = 0$. So $\phi(\eta, a' + \underline{a}) = 0$ makes the household problem a square system for Newton's method.

Three ways this is handled

1. Solve the unconstrained Euler equation, then set $a' = \max\{-\underline{a}, a'_{\text{unc}}\}$ (S6).
2. FB as an equation inside a Newton solve (S4).
3. FB as a *penalty* $\lambda \sum \phi_\varepsilon(\cdot)^2$ — how a neural solver enforces what it cannot impose (S8).

$\phi_\varepsilon = x + y - \sqrt{x^2 + y^2 + \varepsilon^2}$ rounds the kink at the origin. ε too large biases the answer off the constraint; too small kills the Newton step. It is a choice you report.

The Income Process Is the Model's Engine

Everything the model says about wealth runs through $\pi(e, e')$. The standard specification is a discretized AR(1) in logs,

$$\log e_{t+1} = \rho_e \log e_t + \sigma_e \varepsilon_{t+1}, \quad \varepsilon \sim N(0, 1),$$

turned into a finite Markov chain (Tauchen 1986; Rouwenhorst 1995 — we build these in S4).

- ρ_e **governs how long a bad draw lasts**. U.S. earnings estimates put ρ_e near 0.95 annually.
- σ_e **governs how bad it gets**.
- Together they set how much wealth a household *needs* to self-insure — and therefore the whole shape of λ .

Persistence is not a detail

Under **i.i.d.** shocks a bad draw lasts one period, so a modest buffer suffices and almost nobody is ever driven to the constraint.

Under **persistent** shocks a bad spell can last years, the required buffer is large, and a real mass of households ends up constrained.

Same equilibrium concept, same code, radically different economics. Persistence is a parameter *you* choose, and it decides whether the model has any constrained households to study. **We put a number on it in S6.**

How Good Is Self-Insurance?

Assets are a **buffer stock**, not a contract: they insure by *substituting across time*, and unevenly.

- **Rich households** are nearly fully insured — a bad draw is absorbed by running down assets.
- **Households near the constraint** are barely insured: consumption tracks income one-for-one.
- So the **MPC** out of a transfer is near 0 at the top and near 1 at the bottom — heterogeneity a representative agent cannot have, and the reason these models are used for fiscal policy.

Back to the comparison

Under complete markets every household has the same MPC, so who receives a transfer does not matter. Here it matters a great deal — the MPC row of the table, now with a mechanism behind it.

Research in Practice

In 2.A3, assets are what let a household absorb a medical bill or fund a business.

Why the First Welfare Theorem Fails I: One Euler Equation, Not Many

Complete markets. A claim to each next-period shock e' , at price $Q(e')$. One first-order condition *per state*:

$$\frac{u'(c^i(e'))}{u'(c^i)} = \frac{Q(e')}{\beta \pi(e'|e)} \quad \text{for every } e'.$$

The right side is the same for everyone, so every household's marginal rate of substitution across states is **equal** — the condition for Pareto optimality.

One bond. One price q , *one* condition:

$$q u'(c^i) = \beta \sum_{e'} \pi(e'|e) u'(c^i(e')) + \eta^i.$$

It pins only the *average* over e' . Since $c^i(e')$ rises with i 's own e' , $u'(c^i(e'))$ is high exactly when i is unlucky: the MRS across states differs from household to household.

The trade that is missing

Two households, two equally likely states:

state 1: i unlucky, j lucky; state 2: the reverse.

With one bond, u'_i is high in state 1 and u'_j in state 2. Let j give ε to i in state 1, and i give ε to j in state 2. Each gains

$$\frac{1}{2} \varepsilon [u'(\text{unlucky}) - u'(\text{lucky})] > 0.$$

The swap is feasible — it only reshuffles — so the equilibrium is **not Pareto optimal**.

The swap is a claim **contingent on one's own shock**: exactly the security the bond economy does not have.

Why the First Welfare Theorem Fails II: Where the Proof Breaks

The theorem. If markets are **complete**, agents are price takers and there are no externalities, every competitive equilibrium is Pareto optimal.

The proof, in three steps.

1. Suppose a feasible $\hat{c}$ Pareto-dominates the equilibrium c^* .
2. Each household chose c^{*i} from its budget set, so anything it prefers must cost more: $p \cdot \hat{c}^i > p \cdot c^{*i}$.
3. Summing, $\hat{c}$ costs more than the endowment — it is not feasible. Contradiction.

Step 2 is what fails. It needs every bundle to have a price. With one bond, the budget set holds only plans the bond can finance. The swap is not rejected as too *expensive*; it cannot be *bought* at all, so optimality says nothing about it.

What we lose: the planner shortcut

In S1 the welfare theorems let us solve the **planner's** problem and read prices off it. Here no planner's problem reproduces the equilibrium.

So we solve the **equilibrium itself**: household problem, distribution, market clearing — the rest of this deck.

This failure is almost by assumption: the planner is allowed contracts the market is not. The sharper question is whether a planner held to the *same* contracts could do better — next slide.

The Constrained Planner: Same Contracts, Still Better

What “constrained” means. The planner has the **same insurance technology** as the market:

- **no transfers contingent on e** — otherwise it would simply implement full insurance;
- households keep their one bond and their budget constraints, at market prices;
- the planner may only choose (or tax and subsidize) how much each household saves.

It limits what transfers can be *conditioned on*, not their size.

The result. Even so, the equilibrium is usually **constrained inefficient**. Diamond (1967); Geanakoplos & Polemarchakis (1986); Dávila, Hong, Krusell & Ríos-Rull (2012).

Why: a pecuniary externality

Each household takes r as given, but together their saving sets r . A lower r transfers resources from lenders to borrowers.

Complete markets: everyone's u' is proportional, so the transfer is welfare-neutral — the externality nets to zero.

Incomplete markets: borrowers are the unlucky, with high u' . The same transfer is now **insurance**, worth something at first order.

In Aiyagari (2.A2) the price moved is also the wage; which way the planner pushes saving depends on who is poor, and in what.

Stationary Equilibrium in the Huggett Economy

Definition

A **stationary recursive competitive equilibrium** is a bond price q^* , a value function v , policies (c, a') , and a measure λ^* on $\mathcal{A} \times \mathcal{E}$ such that

1. given q^* , (v, c, a') solve the household problem;
2. the bond market clears: $\int a'(a, e) d\lambda^*(a, e) = 0$;
3. λ^* is invariant under the transition induced by $a'(\cdot)$ and π :

$$\lambda^*(\mathcal{B}, e') = \sum_e \pi(e, e') \int_{\{a : a'(a, e) \in \mathcal{B}\}} d\lambda^*(a, e).$$

Why *this* equilibrium?

Stationary — λ constant, prices are *numbers* (2.A, S6).

Recursive / global — Γ moves, prices are *functions* of Γ (2.B1, S6–S8).

Transition path — a sequence $\{\lambda_t\}$ (S6).

Only in the first does the distribution **drop out of the household's state**. Huggett's own reason for stopping there: in 1993 “general methods for characterizing equilibria of the more general kind do not currently exist.”

Why the Interest Rate Must Be Below ρ

The argument, in three steps.

1. Let $\rho \equiv 1/\beta - 1$, the rate of time preference; since $q = 1/(1+r)$, $q = \beta$ means $r = \rho$. Suppose so. Then the Euler equation reads $u'(c_t) = \mathbb{E}_t[u'(c_{t+1})] + \eta_t$, so $\{u'(c_t)\}$ is a **non-negative supermartingale**.
2. A non-negative supermartingale converges almost surely. But with a persistent, non-degenerate endowment and a finite debt limit, $u'(c_t)$ cannot settle down unless assets grow without bound.
3. Every household therefore wants to accumulate forever: aggregate asset demand is $+\infty$. That cannot clear a market in **zero net supply**.

Hence $q^* > \beta$, i.e. $r^* < \rho$.

Read it as a price. A low r is the price that **chokes off** precautionary demand until borrowing equals lending. The more risk, or the tighter the limit, the lower r^* must go.

Research in Practice

The incomplete-markets answer to the risk-free rate puzzle, and why governments borrow cheaply: households cannot insure. Bewley (1986); Huggett (1993); Aiyagari (1994); Chamberlain & Wilson (2000).

The same logic for a government: without commitment, steady-state debt is positive when the sovereign's rate of time preference exceeds the rate on debt; precautionary motives shape its level. Feng & Santos (2026), "The National Debt in a Framework of Time-Consistent Fiscal Policy," WP.

Does a Stationary Equilibrium Exist?

The half that is a theorem

Fix q . If

1. $e_{\min} - (1 - q)\underline{a} > 0$ — at the debt limit, drawing the worst endowment for ever, consumption is still *positive*;
2. $\beta < q$, that is $r < \rho$ — the previous slide;
3. $\pi(e_h | e_h) \geq \pi(e_h | e_l)$ — the shock chain is *monotone*;

then there is a **unique** stationary measure $\lambda(q)$, and iterating any initial distribution forward converges weakly to it — from **any** starting point.

Huggett (1993), Thm. 2, via Hopenhayn & Prescott (1987), Thm. 2 — the *monotone mixing condition*. Those three hypotheses are the three properties of 2.A3, bundled.

The half that is not

None of that delivers the *price*. Huggett is explicit:

"It is not known whether stationary equilibria are unique. If the excess demand for credit balances is continuous and strictly decreasing in the price ... the intermediate value theorem would give the existence of a unique equilibria. These properties have been difficult to prove."

Excess demand was strictly decreasing in every case he computed. The gap is still open — and reappears for Aiyagari's r^* in 2.A3.

What the Model Delivers, and What It Does Not

Delivers

- A non-degenerate **wealth distribution** from identical households and luck alone.
- **Social mobility**, at a rate set by π .
- A risk-free rate **below** the rate of time preference.
- A welfare cost of idiosyncratic risk.
- A stationary distribution that **exists, is unique, and is reached from any start** (Huggett) — and in 2.A2 you watch it happen, in Feng, Han, Sargent & Zhu (2026).

Does not deliver

- The **top tail**. Realistic earnings risk produces nothing like the top 1% wealth share — numbers in 2.A3.
- A theory of *why* markets are incomplete — that is the other two roads.
- Any role for aggregate risk — prices are constant by construction (2.B1).
- Finite lives, bequests, retirement (2.B2).

- Full insurance is a **testable** prediction about consumption growth. It is rejected.
- Assume one asset and a limit and you get precautionary saving, a wealth distribution and $r^* < \rho$ from identical households — with the equilibrium object a **measure** that must reproduce itself.

Bridge: Put a Firm in the Room

Huggett's bond is in **zero net supply**: one household's asset is another's debt, and the interest rate clears that internal market.

Now let the asset be a claim on **physical capital**.

Then:

- aggregate saving is no longer zero — it is the capital stock K ;
- K has a **price determined by technology**, through the firm's first-order condition;
- the wage appears, so the endowment becomes labour income $w\epsilon$.

Next (2.A2): Aiyagari

The same household problem, closed with a neoclassical firm. Equilibrium becomes the crossing of a **capital demand** curve from the firm and a **capital supply** curve from the distribution of households.

And the first quantitative disappointment of the course.